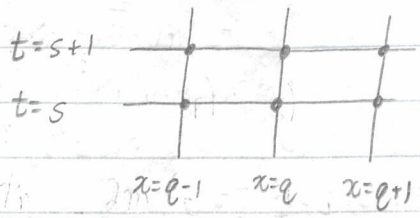
Crank Nicolson Method

- We ultimately want to solve the Time-Dependent Schrödinger Equation:

$$i\hbar \frac{\delta \Psi}{\delta t} = \hat{H}\Psi = -\frac{\hbar^2}{2m}\frac{\delta^2 \Psi}{\delta x^2} + V(x)\Psi$$

- But, to allow our computer to begin to compute the derivatives, we must define a grid:



> At $x=q, t=s$, we can approximate: $\frac{\delta^2\Psi}{\delta x^2} \approx \frac{1}{(\Delta x)^2}\left(\Psi(x=q+\Delta x, t=s) - 2\Psi(x=q, t=s) + \Psi(x=q-\Delta x, t=s)\right)$

> At $x=q, t=s+\Delta t$, we can approximate: $\frac{\delta^2\Psi}{\delta x^2} \approx \frac{1}{(\Delta x)^2}\left(\Psi(x=q+\Delta x, t=s+\Delta t) - 2\Psi(x=q, t=s+\Delta t) + \Psi(x=q-\Delta x, t=s+\Delta t)\right)$

> Since we're taking a small time step, using only the approximation for $\frac{\delta^2\Psi}{\delta x^2}\Big|_{t=s}$ or only the approximation for $\frac{\delta^2\Psi}{\delta x^2}\Big|_{t=s+1}$ won't be particularly accurate. Instead, we will average the two approximations:

$$\frac{\delta^2\Psi}{\delta x^2} \approx \frac{1}{2(\Delta x)^2}\left(\Psi(q+\Delta x, s) - 2\Psi(q,s) + \Psi(q-\Delta x, s) + \Psi(q+\Delta x, s+\Delta t) - 2\Psi(q, s+\Delta t) + \Psi(q-\Delta x, s+\Delta t)\right)$$

> Using similar logic, we can approximate Ψ on the right-hand side of the TDSE as:

$$\Psi \approx \frac{1}{2}\left(\Psi(q,s) + \Psi(q, s+\Delta t)\right)$$

> Finally, we can approximate $\frac{\delta\Psi}{\delta t}$ at $x=q, t=s$ as:

$$\frac{\delta\Psi}{\delta t} \approx \frac{\Psi(q, s+\Delta t) - \Psi(q,s)}{\Delta t}$$

- We can plug our approximations in to see, at $x=q$:

$$i\hbar \cdot \frac{\Psi(q,s+\Delta t) - \Psi(q,s)}{\Delta t} = -\frac{\hbar^2}{2m}\cdot\frac{1}{2(\Delta x)^2}\left(\begin{array}{l}\Psi(q+\Delta x, s) - 2\Psi(q,s) + \Psi(q-\Delta x, s) \\ + \Psi(q+\Delta x, s+\Delta t) - 2\Psi(q, s+\Delta t) + \Psi(q-\Delta x, s+\Delta t)\end{array}\right) + \frac{V(q)}{2}\left(\begin{array}{l}\Psi(q,s) \\ + \Psi(q, s+\Delta t)\end{array}\right)$$

$$\Rightarrow \Psi(q,s+\Delta t) - \Psi(q,s) = -\frac{i\Delta t}{2\hbar}\left[-\frac{\hbar^2}{2m(\Delta x)^2}\left(\begin{array}{l}\Psi(q+\Delta x, s) - 2\Psi(q,s) + \Psi(q-\Delta x, s) \\ + \Psi(q+\Delta x, s+\Delta t) - 2\Psi(q, s+\Delta t) + \Psi(q-\Delta x, s+\Delta t)\end{array}\right) + V(q)\left(\begin{array}{l}\Psi(q,s) \\ + \Psi(q, s+\Delta t)\end{array}\right)\right]$$

- Our goal is to have $\Psi(*, s+\Delta t)$ on one side of the equation, and $\Psi(*, s)$ on the other:

$$\Psi(q,s+\Delta t) + \frac{i\Delta t}{2\hbar}\left[-\frac{\hbar^2}{2m(\Delta x)^2}\left(\Psi(q+\Delta x, s+\Delta t) - 2\Psi(q, s+\Delta t) + \Psi(q-\Delta x, s+\Delta t)\right) + V(q)\Psi(q, s+\Delta t)\right]$$
$$= \Psi(q,s) - \frac{i\Delta t}{2\hbar}\left[-\frac{\hbar^2}{2m(\Delta x)^2}\left(\Psi(q+\Delta x, s) - 2\Psi(q,s) + \Psi(q-\Delta x, s)\right) + V(q)\Psi(q,s)\right]$$

- We'll first simplify this by writing $\alpha = \frac{i\hbar\Delta t}{4m(\Delta x)^2}$ and $\beta = \frac{i\Delta t}{2\hbar}$, so:

$\Psi(q, s+\Delta$

$\Psi(q, s+\Delta t) = \alpha\Psi(q+\Delta x, s+\Delta t) + 2\alpha\Psi(q, s+\Delta t) - \alpha\Psi(q-\Delta x, s+\Delta t) + \beta V(q)\Psi(q, s+\Delta t)$

$= \Psi(q,s) + \alpha\Psi(q+\Delta x, s) - 2\alpha\Psi(q, s) + \alpha\Psi(q-\Delta x, s) - \beta V(q)\Psi(q,s)$

$\Rightarrow (1+2\alpha+\beta V(q))\Psi(q, s+\Delta t) - \alpha\Psi(q+\Delta x, s+\Delta t) - \alpha\Psi(q-\Delta x, s+\Delta t)$

$= (1-2\alpha-\beta V(q))\Psi(q, s) + \alpha\Psi(q, s) + \alpha\Psi(q, s)$

- This creates a system of equations. For simplicity have the minimum values of q to be zero, and we shall let $\Delta x = 1$ for simplicity. We will take $\Psi(-1, s+\Delta t) = 0$ and $\Psi(n+1, s+\Delta t) = 0$ at the boundary (and similarly, $\Psi(-1,s) = \Psi(n+1,s) = 0$)

$$(1+2\alpha+\beta V(0))\Psi(0, s+\Delta t) - \alpha\Psi(1, s+\Delta t) = (1-2\alpha-\beta V(0))\Psi(0,s) + \alpha\Psi(1,s)$$

$$-\alpha\Psi(0, s+\Delta t) + (1+2\alpha+\beta V(1))\Psi(1, s+\Delta t) - \alpha\Psi(2, s+\Delta t) = \alpha\Psi(0,s) + (1-2\alpha-\beta V(1))\Psi(1,s) + \alpha\Psi(1,s)$$

$$-\alpha\Psi(1, s+\Delta t) + (1+2\alpha+\beta V(2))\Psi(2, s+\Delta t) - \alpha\Psi(3, s+\Delta t) = \alpha\Psi(1,s) + (1-2\alpha-\beta V(2))\Psi(2,s) + \alpha\Psi(3,s)$$

$$\vdots$$

$$-\alpha\Psi(n-2, s+\Delta t) + (1+2\alpha+\beta V(n-1))\Psi(n-1, s+\Delta t) - \alpha\Psi(n, s+\Delta t) = \alpha\Psi(n-2,s) + (1-2\alpha-\beta V(n-1))\Psi(n-1,s) + \alpha\Psi(n,s)$$

$$-\alpha\Psi(n-1, s+\Delta t) + (1+2\alpha+\beta V(n))\Psi(n, s+\Delta t) = \alpha\Psi(n-1,s) + (1-2\alpha-\beta V(n))\Psi(n,s)$$

> I have let n equal the total number of partition points of the x interval.

- Note that $\Psi(*, s)$ are all considered to be known. We can decide to let: $\heartsuit_0 = (1-2\alpha-\beta V(0))\Psi(0,s) + \alpha\Psi(1,s)$, $\heartsuit_1 = \alpha\Psi(0,s) + (1-2\alpha-\beta V(1))\Psi(1,s) + \alpha\Psi(1,s)$, ... $\heartsuit_j = \alpha\Psi(j-1,s) + (1-2\alpha-\beta V(j))\Psi(j,s) + \alpha\Psi(j+1,s)$. All of the $\heartsuit$'s are known values.

- We can also note that our system of equations can be written as a matrix equation:

$$\begin{pmatrix} 1+2\alpha+\beta & -\alpha & 0 & 0 & 0 & \cdots & 0 \\ -\alpha & 1+2\alpha+\beta & -\alpha & 0 & 0 & \cdots & 0 \\ 0 & -\alpha & 1+2\alpha+\beta & -\alpha & 0 & \cdots & 0 \\ 0 & 0 & -\alpha & 1+2\alpha+\beta & -\alpha & \cdots & 0 \\ 0 & 0 & 0 & -\alpha & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & -\alpha \\ 0 & 0 & 0 & \cdots & \cdots & -\alpha & 1+2\alpha+\beta \end{pmatrix} \begin{pmatrix} \Psi(0, s+\Delta t) \\ \Psi(1, s+\Delta t) \\ \Psi(2, s+\Delta t) \\ \Psi(3, s+\Delta t) \\ \Psi(4, s+\Delta t) \\ \vdots \\ \Psi(n, s+\Delta t) \end{pmatrix} = \begin{pmatrix} \heartsuit_0 \\ \heartsuit_1 \\ \heartsuit_2 \\ \heartsuit_3 \\ \heartsuit_4 \\ \vdots \\ \heartsuit_n \end{pmatrix}$$

- This matrix equation can be solved by using the Thomas Algorithm. This is done by:

① Divide the first row (equation) by $1+2\alpha+\beta$

First row only $\longrightarrow \begin{pmatrix} 1 & \frac{-\alpha}{1+2\alpha+\beta} & 0 & 0 & \cdots & 0 \\ & \vdots & & & & \end{pmatrix} = \begin{pmatrix} \Psi(0, s+\Delta t) \\ \vdots \end{pmatrix} = \begin{pmatrix} \frac{\heartsuit_0}{1+2\alpha+\beta} \\ \vdots \end{pmatrix}$

② Add α times the first row to the second row:

only 1st and second rows changed

$$\begin{pmatrix} 1 & -\frac{\alpha}{1+2\alpha+\beta} & 0 & \cdots & 0 \\ 0 & 1+2\alpha+\beta-\frac{\alpha^2}{1+2\alpha+\beta} & \alpha & \cdots & 0 \\ & \vdots & & & \end{pmatrix} \begin{pmatrix} \Psi(0, s+\Delta t) \\ \vdots \end{pmatrix} = \begin{pmatrix} \frac{\nabla_0}{1+2\alpha+\beta} \\ \nabla_1 + \frac{\alpha \nabla_0}{1+2\alpha+\beta} \\ \vdots \end{pmatrix}$$

③ Repeat steps 1 & 2, dividing by the "middle" coefficient to make the main diagonals all equal to 1, and subtracting by $\alpha \cdot (\text{row})$ to ensure the final matrix is upper triangular.

④ We can write the final matrix as:

$$\begin{pmatrix} 1 & F_0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & F_1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 1 & F_2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & F_n \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} \Psi(0, s+\Delta t) \\ \Psi(1, s+\Delta t) \\ \Psi(2, s+\Delta t) \\ \Psi(3, s+\Delta t) \\ \vdots \\ \Psi(n, s+\Delta t) \end{bmatrix} = \begin{pmatrix} d_0 \\ d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_n \end{pmatrix}$$

where:

$$F_0 = -\frac{\alpha}{1+2\alpha+\beta}, \quad F_1 = \frac{-\alpha}{1+2\alpha+\beta+\alpha F_0}, \quad F_j = \frac{-\alpha}{1+2\alpha+\beta+\alpha F_j}$$

$$d_0 = \frac{\nabla_0}{1+2\alpha+\beta}, \quad d_1 = \frac{\nabla_1 + \alpha d_0}{1+2\alpha+\beta+\alpha F_0}, \quad d_j = \frac{\nabla_j + \alpha d_{j-1}}{1+2\alpha+\beta+\alpha d_j}$$

⑤ Once we have our finalised matrix in part #4, we can backsubstitute to have:

$$\Psi(n, s+\Delta t) = d_n$$

$$\Psi(n-1, s+\Delta t) + F_n \Psi(n, s+\Delta t) = d_{n-1} \Rightarrow \Psi(n-1, s+\Delta t) = d_{n-1} - F_n \Psi(n, s+\Delta t)$$

⑥ In general, we can say $\Psi(j, s+\Delta t) = d_j - F_j \Psi(j+1, s+\Delta t)$ where $\Psi(n, s+\Delta t) = d_n$

- From here, all the math has been done and all that's left is to implement this!